



AIM-355
ACCELERATED CURING TANK



Aimil Ltd.
Instrumentation & Technologies



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AIM - 355-1

AIM - 355-2

AIM - 355-3

AIM - 355-4

AIM - 355-1-ACW

AIM - 355-2-ACW

AIM - 355-1-ACB

AIM - 355-2-ACB

Accelerated Curing Tank

Operating Instructions



Accelerated Curing Tank

The equipment comprises of the following :

1. One Double walled chamber with stainless steel walls, fitted with a water circulating pump, heating elements and panel a fitted with a single phase indicating lamp (CONT) one digital temperature controller, a limit switch, an RCCB mains switch and a Heater Switch.
2. Sample supporting tray – as per table A.
3. Top Lid .
4. One copy of Operating Instructions with General Assembly Drawing

Caution

The equipment is fitted with heaters of different specifications as show in the table below.

Table A

Cat. No.	Chamber Specification			Heater Specification		
	SS Trays	No. of cube moulds 150MM/ 70.6MM	Size of Chamber LxBxH	Power (KW)	Operating Voltage (V)	Current Rating (Amperes)
AIM-355-1	1	6/12	22"x15"x13"	4.0	220	17.0
AIM-355-2	2	12/24	22"x15"x20"	4.0	220	17.0
AIM-355-3	2	24/48	29"x22"x20"	4.5	440	19.0
AIM-355-4	2	36/72	43"x22"x20"	6.0	440	25.5
AIM-355-1-ACB	1	06/12	40"x22"x20"	6.0	440	25.5
AIM-355-2-ACB	2	12/24	40"x22"x28"	9.0	440	38.25
AIM-355-1-ACW	1	06/12	40"x22"x15"	4.5	440	19.0
AIM-355-2-ACW	2	12/24	40"x22"x24"	6.0	440	25.5

220V = Single phase

440V = Three phase



Operating Instructions **Accelerated Curing Tank**

Introduction

The quality of concrete is traditionally calculated in terms of its 28 day's compressive strength by 28 days of moist curing before testing. This period is too long a period to be of any value for either concrete construction control or applying timely corrective measures. For assessing the quality of controlled concrete, an acceptance test which can supply results, while the concrete is still accessible and sufficiently green to make its removal practicable; that is, within about 24 hours after casting is needed. With the assistance of reliable test methods employing the accelerated curing technique, it is now possible to test the compressive strength of concrete within a short period and thereby to estimate whether it is likely to reach the specified strength at 28 days or not.

The Accelerated Curing Tank /Curing Tank is designed for curing of specimens of concrete, under conditions intended to accelerate the development of strength, as described in IS:9013.

The Accelerated Curing Tank is used for curing moulds in fresh water at the desired uniform temperature for stipulated time periods, until taken out for testing.

The machine is suitable for operation on a 415V, Three Phase, AC Supply.

Description

The accelerated Curing Tank / Curing Tank is illustrated in the General Assembly Drawing attached. The tank(01) is an insulated tank, with a top lid / cover (03). The samples are kept inside the tank for curing. The handle (02) is used for opening or closing the lid. Power to the curing tank is supplied using the mains lead (04). A microswitch (10) is fitted in heater control circuit disconnects the supply to the heater whenever the lid(03) is opened. Supply to the heaters (16) is resumed when the lid(03) is closed.

Heaters(16) are mounted inside the tank(1). A water circulation pump (17) circulates the water in the tank (1) continuously to ensure uniform temperature is maintained in the tank. After the test is over, the water in the tank is drained through the drain valve (14). The temperature of the curing tank can be programmed to a value set by the user using the PID temperature controller/Indicator (05) fitted to the front panel. The front panel also contain the Mains Switch (07), the heater "ON" power indicator(06) which glows whenever the heater is energized. Also, there is a heater control switch (08) and an RCCB (09) which protects the operator against leakage current.

The tank(01) and the lid(03) are interfaced via. a silicon rubber gasket(11), which seals the steam from escaping during the curing process. Leveling bolts (12) are provided at the four bottom corners of the tank, using which the complete curing tank can be leveled on the flooring using a user supplied spirit level. There is a stopper arm(13), which can be used to hold the lid(03) during the loading / unloading of the sample.



Setting Up And Operation

Place the unit on firmly leveled ground. Connect the mains lead (4) of the unit to a 415V, 3- Phase, A.C. Supply. Switch the unit on by operating the Main switch and Heater switch.

Temperature can be adjusted and indicated by the temperature controller (5). Press the SET key adjust the desired temperature. The temperature can then be increased or decreased by the ▲ or ■ keys. Then press the ENT key. The temperature will now move towards the set temperature.

Connect the inlet tap to a clean potable water supply and open the tap to fill the tank, so that both the sample trays should be dipped in water.

The specimens to be cured shall be prepared in accordance with the guidelines given in IS:516.

Accelerated Curing by Warm Water Method

After the specimens have been made, they shall be left to stand undisturbed in their moulds in a place free from vibration at a temperature between 25°C to 29°C for at least one hour, prior to immersion in the curing tank. The time between the addition of water to the ingredients and immersion of the test specimens in the curing tank shall be at least 1½ hours, but shall not exceed 3½ hours.

Adjust the temperature controller switch (6) and maintain the temperature of the water inside the tank at $55 \pm 2^\circ\text{C}$.

The specimens in their moulds shall be gently lowered into the curing tank and shall remain totally immersed in water at $55 \pm 2^\circ\text{C}$ for a period of not less than 19 hours 50 minutes. The specimens shall then be removed from the water, marked for identification, removed from their moulds and immersed in the cooling tank at a temperature of $27 \pm 2^\circ\text{C}$ before the completion of 20 hours 10 minutes from the start of immersion in the curing tank. They shall remain in the cooling tank for a period of not less than one hour.

Note: For control purposes, the above time tolerances may be relaxed & an appropriate correction factor applied.

The specimens shall then be tested in accordance with IS:516. In the warm water method, specimens shall be tested while still set, not more than 2 hours from the time of immersion in the cooling tank. In the boiling water method, the age at the time of test shall be 28.5 hours $\pm$ 20 minutes.



Accelerated Curing by Boiling Water Method

After the specimens have been made, they shall be stored in a place free from vibration, in moist air of at least 90 percent relative humidity and at a temperature of $27 \pm 2^\circ\text{C}$ for 23 hours $\pm$ 15 minutes from the time of addition of water to the ingredients.

The specimens shall then be gently lowered into the curing tank and shall remain totally immersed for a period of $3\frac{1}{2}$ hours $\pm$ 5 minutes. The temperature of the water in the curing tank shall be at boiling (100°C) at sea level. The temperature of water shall not drop more than 3°C after the specimens are placed and shall return to boiling within 15 minutes.

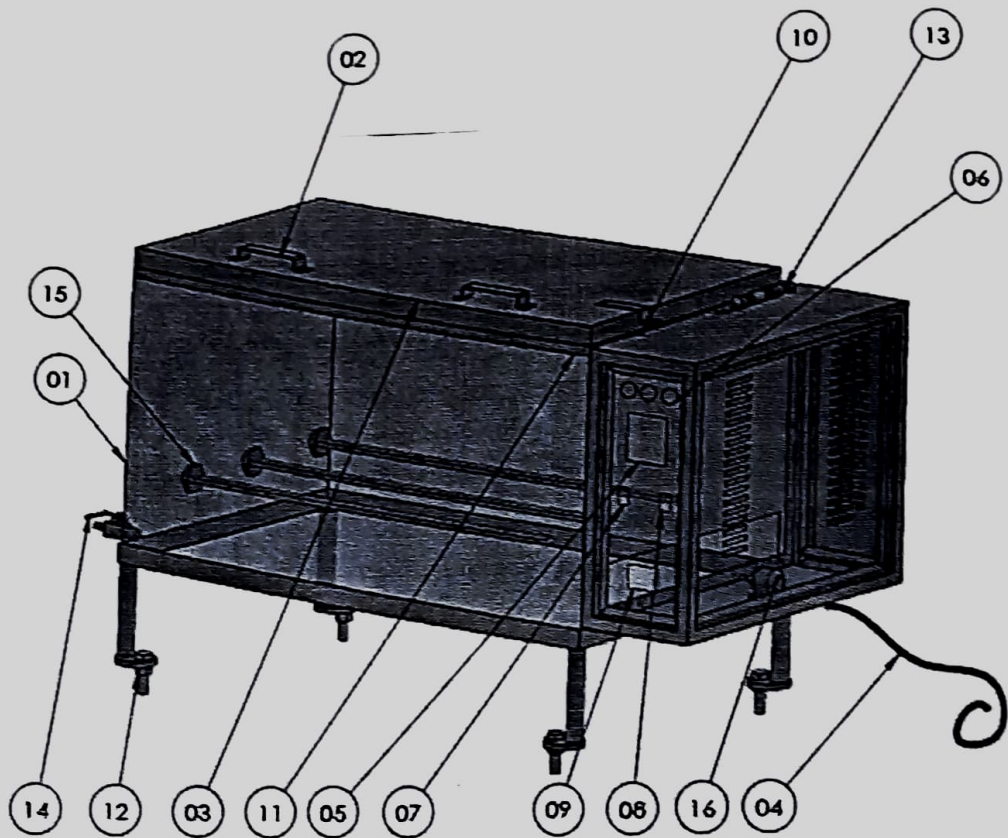
Note: In confined places, the temperature of the water may be kept just below the boiling point to avoid excessive evaporation.

After curing for $3\frac{1}{2}$ hours $\pm$ 5 minutes in the curing tank, the specimen shall be removed from the boiling water, removed from the moulds and cooled by immersing in the cooling tank at $27 \pm 2^\circ\text{C}$ for 2 hours.

Maintenance

1. The equipment must be cleaned and dried before storage.
2. After the test is over all the water from the chamber, must be drained and the chamber must then be thoroughly cleaned and dried with a cloth duster.

For Three Phase Supply



GA DRAWING CURING TANK
AIM- 355-3,355-4,355-1ACB, 355-
2ACB, 355-1ACW, 355-2ACW,

S. No.	DESCRIPTION
01	CABINET
02	HANDLE
03	LID
04	MAINS LEAD
05	TEMPERATURE INDICATOR CONTROLLER
06	SUPPLY INDICATOR
07	MAINS SWITCH
08	HEATER SWITCH



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S. No.	DESCRIPTION
09	RCCB
10	MICROSWITCH
11	GASKET
12	LEVELLING BOLT
13	DOOR STOPPER ARM
14	DRAIN VALVE
15	HEATERS
16	CIRCULATING PUMP